

Transformer Based Cross Resolution Feature Fusion in Ensemble U-Net for Landslide Detection

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Abstract

Landslides are sudden and high-impact hazards that demand rapid and reliable detection. This study presents a segmentation framework that integrates an extended U-Net backbone with transformer blocks and coordinate attention to capture long-range dependencies while maintaining fine terrain detail. Cross-resolution feature fusion enhances boundary precision for small landslide regions, and Tversky-based loss mitigates class imbalance during training. The model utilises multi-spectral inputs informative for geomorphological interpretation, including RGB, NDVI, SWIR, slope and elevation. A refined preprocessing stage improves input separability through LAB-space CLAHE for local contrast enhancement, followed by edge fusion using Canny, Laplacian of Gaussian and Scharr operators to emphasise structural features. An NDVI-based vegetation mask suppresses redundant canopy edges, while selective unsharp masking and bilateral filtering enhance terrain contrast without introducing noise. The resulting eight-channel feature stack comprises enhanced RGB, NDVI, slope, elevation, SWIR1 and SWIR2 bands. Evaluated on the Landslide4Sense dataset, the system achieved an F1 score of 0.938, mean IoU of 0.835 and accuracy of 0.995, demonstrating strong generalisation across diverse terrains. Its transformer-enhanced architecture and vegetation-aware preprocessing make it suitable for large-scale mapping, real-time early warning and on-site disaster risk assessment.

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1. Introduction

Landslides represent one of the most devastating types of natural events characterized as extensive destruction of human life, property, and nature in general, rapidly occurring in areas with a high gradient and intensive precipitation. Given urbanization tendencies and extensive construction on the slopes that are susceptible to landslides, identifying and mapping landslides is of paramount importance today. Traditional monitoring approaches, including field surveys and remote sensing, are insufficiently efficient, as they are laborious, expensive, and hardly applicable for long-time monitoring. Due to the above, the use of satellite sensing integrative with deep learning is a novel trend for landslide instability evaluation.

The main objective of this work is to design a sophisticated landslide detection system using a transformer-infused U-Net architecture and multi-spectral and parametric input. As opposed to the conventional system that relied only on the quasi-raw RGB data, this latter integrates numerous derived indices including NDVI, slope, elevation, and SWIR. Such a system captures both the superficial and sub-superficial features of the target area relevant to the landslide. Additionally, the system also incorporated a more sophisticated preprocessing step, which includes LAB-space CLAHE, edge fusion, vegetation mask, and bilateral filter, that are intended to enhance the characteristics of such features and with the aim of reducing the ambiguous nature of the noisy one. This is believed to improve the accuracy of segmentation across different terrain types while maintaining the computationally inexpensive nature of the model, which is suitable for both huge-scale map production and practical use in the disaster-prone area itself.

2. Related Works

2.1. History

Early work in landslide detection mainly depended on manual interpretation of aerial photographs, field surveys and topographic inspection. These methods were useful for understanding slope failures but were slow, labour intensive and unsuitable for large scale or repeated monitoring. With the growing availability of satellite remote sensing, researchers shifted to analysing multi spectral images using threshold rules, texture measures and basic statistical techniques. These approaches marked an important step forward but still struggled in areas with dense vegetation, complex terrain shapes or seasonal changes in reflectance.

As computational methods advanced, machine learning and later deep learning models became central to this field. Initial studies used handcrafted features derived from NDVI, slope, elevation and soil exposure metrics. However, the introduction of encoder decoder architectures like U Net significantly improved pixel level segmentation and established the basis for modern landslide mapping. Recent research, including works such as [3], demonstrates the effectiveness of combining multi spectral information with attention based or transformer enhanced networks. These developments have led to more reliable detection of small, fragmented and subtle landslide regions, setting the direction for current hybrid deep learning approaches.

2.2. Existing Approaches

Research on landslide detection has steadily moved from classical image processing techniques to deep learning based segmentation models. Early studies used thresholding, band ratio rules and texture measures to identify disturbed soil and exposed surfaces, but these methods struggled under varying lighting and vegetation conditions. With the availability of annotated datasets such as Landslide4Sense introduced by Ghorbanzadeh et al. [1], deep models became more practical for large scale landslide mapping. This was later expanded through further benchmarking work by Schlögl et al. [2], which provided a strong comparative foundation for evaluating segmentation networks. Another important contribution is the CAS Landslide Dataset by Xu et al. [3], which introduced multi sensor and multi temporal information and encouraged the development of generalisable models capable of handling varied terrain types.

Among the deep learning models explored in literature, U Net variants and attention based architectures have shown strong performance in segmenting small and fragmented landslide regions. Several works experimented with spectral fusion, edge based enhancement and transformer modules to improve boundary clarity, although not all of them generalise well across different landscapes. The most relevant work to the present study is the Inception based U Net model described in the base paper [4], which demonstrated that multi branch convolution and improved feature extraction significantly boost segmentation accuracy in remote sensing tasks. Along with this, other transformer based and attention guided studies are also discussed in recent literature, but their contributions are referenced only broadly here since the main focus is on the architectural ideas that influence the proposed system. Overall, these existing approaches form the basis for developing hybrid CNN transformer models with enhanced preprocessing and cross resolution fusion to achieve more stable and precise landslide detection.

2.3. Research Gap

Although existing deep learning models have shown considerable improvement in landslide segmentation, several limitations remain that directly influence real world applicability. Many current approaches perform well on clear and exposed slopes but struggle in areas with dense vegetation, shadow effects or mixed land cover. This reduces their ability to generalise across regions with varied geological conditions. Even models trained on large datasets often fail to accurately capture very small or early stage landslide patches, which are important indicators for early warning systems. Another limitation arises from class imbalance, where landslide pixels occupy only a small portion of an image, causing models to predict background regions more confidently than actual slide areas.

A second gap relates to boundary precision and interpretability. While recent architectures like the Inception based U Net [4] have improved feature extraction, many models still produce blurred or incomplete boundaries, especially in rugged terrain. Transformer based studies help with global context, but they often require extensive computational resources and are sensitive to noise in multi spectral inputs. Existing works also provide limited explainability, making it difficult to understand how individual spectral channels influence the prediction outcome. Furthermore, most deep learning studies produce only binary segmentation masks and do not provide an interpretable risk assessment layer. This creates a gap between pixel level predictions and actionable decision making. The present work addresses these issues by incorporating cross resolution feature fusion, transformer enhancement and a fuzzy inference module to generate more accurate and interpretable outputs across diverse landscapes.

3. Dataset Description

3.1. Dataset Exploration

The landslide4sense dataset is a collection of satellite images. That's going to support the model, since in a real-world scenario it has to deal with terrains and conditions in as wide a range as possible. For that reason, the dataset was designed under high-quality imagery sources, such as Sentinel-2 and ALOS PALSAR, whose purpose is the contribution to landslide detection research. Its rich and varied data make it a good source for machine learning and remote sensing studies.

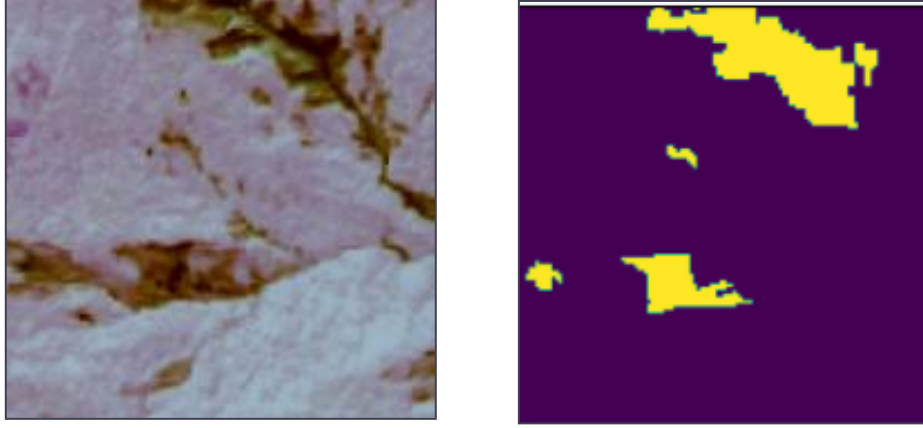


Figure: 1 Landslide Image and its Mask

The above two images form the dataset. The first is a captured landslide area by a satellite, emphasizing the natural features and variations of the terrain. In the second image, the segmentation mask is shown and highlighted in yellow for the landslide regions and purple for the background. Such images would be helpful for the model to understand and learn how to distinguish landslide areas from the general terrain.

3.2. Image Format And Spectral Bands

The Landslide4Sense dataset has images in HDF5 (H5) format, with each containing 14 spectral bands from different satellite sensors. The bands capture unique wavelengths of light which are rich in information to distinguish landslide areas from the background. The table below describes the included spectral bands and their respective wavelengths are as follows:

Table 1: Spectral band and Wavelength Ranges

Spectral Band	Coastal	Blue	Green	Yellow	Red	Red Edge	NIR 1
Wavelength (nm)	400–450	450–510	585–625	630–680	705–745	770–895	770–895
Spectral Band	NIR 2	SWIR 1	SWIR 2	SWIR 3	TIR 1	TIR 2	Panchromatic
Wavelength (nm)	860–1040	1230–1250	1550–1750	2080–2350	10400–12500	10400–12500	450–800

3.3. Dataset Composition

The dataset comprises:

- **Total Images:** 3,799 images in H5 format, where every image is associated with a binary-segmented ground truth mask.
- **Images with Landslides:** 2,231 images contain visible landslides.
- **Background Images:** 1,568 images have no landslides; only background terrain can be seen.

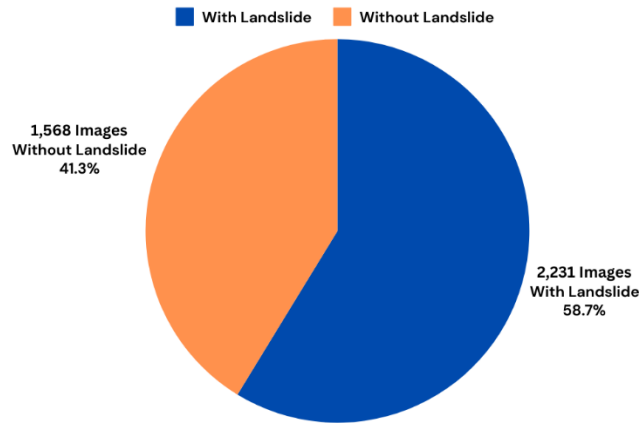


Figure 2: Class Distribution of Dataset

Given that over 40% of images do not have landslide, we do have class imbalance problem, and our model may be biased to classify this image as no landslide so as not to misclassify any segment hence for that under sampled images with no landslide, we should perform some augmented sampling on images of landslide. The images were pre-processed into the following 8 spectral bands : **RGB, NDVI, slope, elevation, SWIR1, SWIR2, and target mask** for training.

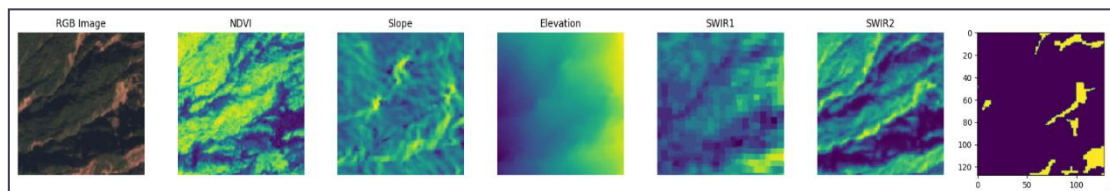


Figure 3: Custom-selected spectral bands extracted for training.

Preprocessing the Landslide4Sense dataset is the first step in our model's workflow, providing clearer and more informative images to the model. In the dataset, the satellite images are not clear and faint, denoting this does not contain highly rich features for predicting landslides. We can pre-process these images such that it becomes easier for the model to learn and will be able

to identify landslide regions more accurately. This step sets a foundation for the entire model by clarifying the images and better emphasizing their features.

4. Design

4.1. System Architecture and Workflow

The proposed system is developed as an end-to-end framework that performs satellite image enhancement, multi-spectral fusion, transformer-driven feature learning and structured post-processing for accurate landslide segmentation. The design relies on a combination of spectral refinement, cross-resolution learning and contextual attention to capture small, irregular terrain deformations that typically indicate slope failure. The workflow begins with a detailed enhancement stage, where each spectral band undergoes percentile normalization, LAB-space CLAHE, NDVI-guided masking, as well as multi-scale edge construction through Canny, LoG and Scharr operators. These steps refine the terrain boundaries and suppress vegetation noise. The enhanced channels are fused into an eight-layer composite consisting of RGB, NDVI, slope, elevation, SWIR1 and SWIR2, which together represent both surface reflectance and geomorphological structure.

This input flows into a hybrid CRFF Transformer U-Net capable of learning both local textures and wider spatial patterns. The encoder extracts hierarchical features while the transformer bottleneck interprets long-range relationships across the terrain. The decoder integrates coordinate attention, skip-connections and edge-refinement blocks to reconstruct sharp boundaries. Class imbalance is addressed through balanced sampling and a combined Dice–Tversky loss.

4.2. Module Descriptions

The overall system is made up of different interconnected modules which are functioning one after another to get the result fast and accurate landslide segmentation. Every unit in the system carries out a particular function that increases the confidence level of the entire system. The journey from data preparation and data enhancement to feature creation is done in such a way that each stage is an improvement in terms of spatial resolution, spectral separability, and interpretability of the dataset. Thanks to the modular design, the system is both scalable and adaptable to different terrains and weather conditions. The main modules are outlined below.

4.2.1. Data Acquisition and Formatting

The focus of this module is on gathering and organizing multi-spectral and terrain-based images that are to be used for training and validating the model. The data are obtained from the high-resolution remote-sensing instruments which are capable of capturing several spectral bands in both optical and infrared ranges. The pre-processing performed at this stage include spectral alignment, noise reduction and standard normalisation of the reflectance values to ensure that the data are uniformly represented. By keeping the spectral quality and spatial resolution consistent, this stage is setting up a solid base for further refinement and model integration.

4.2.2. Pre-Processing and Edge-Aware Enhancement

Pre-processing is a key ingredient in the recipe of making the images clearer and the fine details of the terrain more visible. The work is initiated by the contrast enhancement in the LAB color space where CLAHE is used to even out the local illumination. The edge features are obtained by performing the fusion of gradient-based filters such as Canny, Laplacian of Gaussian and Scharr at multiple scales thus allowing the system to identify not only the weak but also the strong edges. The NDVI-driven vegetation mask is used to eliminate the noise that comes from the areas with the thick canopy, whereas the selective sharpening and bilateral smoothing are used to keep the natural texture of the slopes intact. The method employed here not only improves the spectral and structural characteristics of the images but also makes the segmentation model get a better grasp of the terrain.

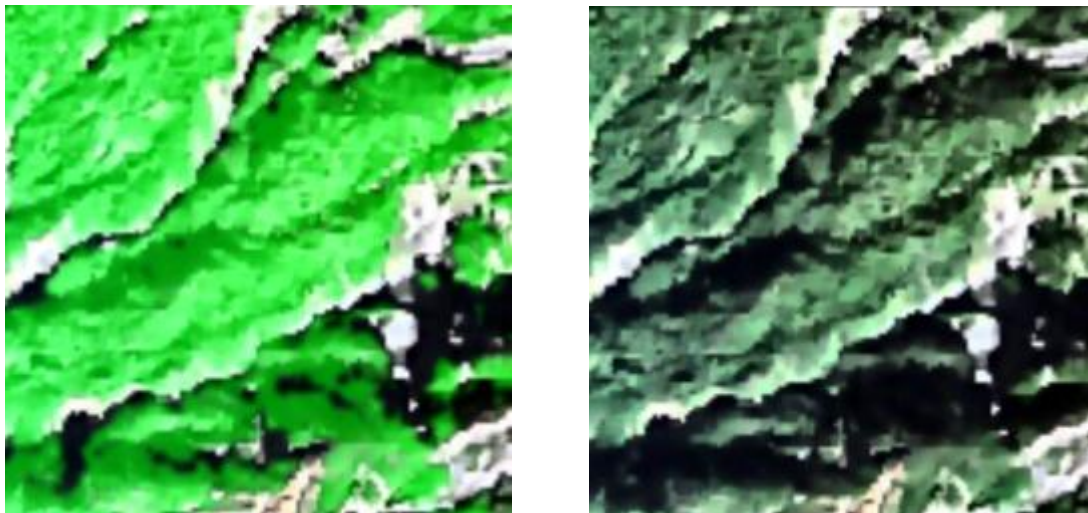


Figure: 4 Vegetation Highlight & Enhanced RGB

4.2.3. Feature Stack Generation

This module is about the integration of the optical and terrain-derived features that have been enhanced into one feature stack. The combined representation most of the time consists of the

visible, infrared and topographic variables that, if taken together, can explain the health of the vegetation, the moisture content in the soil and the morphology of the slope. By combining these complementary features, the system has a complete understanding of the surface dynamics that leads to slope instability. Such a rich, multi-dimensional input enables the deep learning model to generalise effectively across diverse landscapes and environmental conditions.

4.2.4. Hybrid CRFF U-Net Architecture

The hybrid encoder–decoder model with an improved U-Net backbone merged with a Cross-Resolution Feature Fusion (CRFF) technique is at the heart of the proposed system. The encoder captures hierarchical features at different spatial resolutions step by step, and the decoder through the skip connections reconstructs detailed segmentation masks. The CRFF method merges the data coming from different scales of feature to make sure that after up sampling both the coarse contextual clues and the fine structural details are there. This invention enhances the model's capability to find tiny landslide scars and keeps the terrain going even in areas with complex textures or irregular slopes. The modular structure of this design facilitates the easy insertion of transformer and attention units for further polishing.

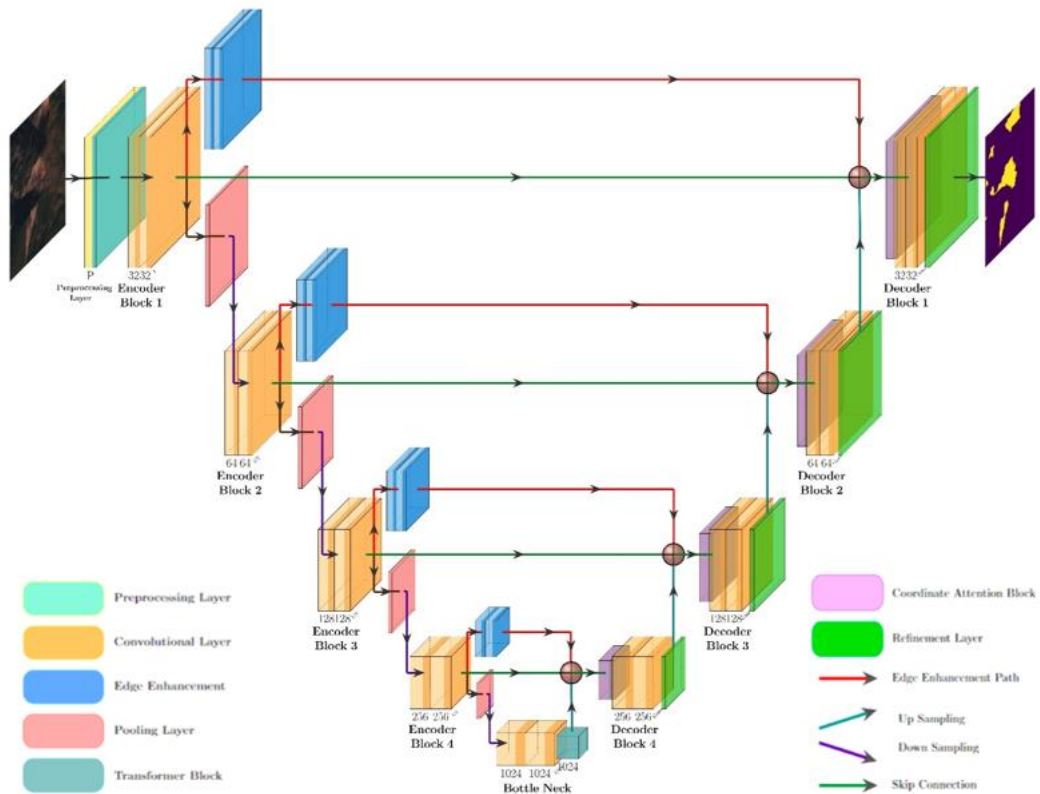


Figure 5: Hybrid CRFF Transformer Based U-Net Architecture

4.2.5 Transformer Bottleneck Module

The transformer module is at the network's bottleneck, where it can most powerfully capture global spatial relationships. While convolutions are good for extracting local features, transformers can capture long-term dependencies which are vital for the comprehension of large-scale terrain changes. Multi-head self-attention is the tool that is used to link the different parts of the image, and this enables the model to find spatial patterns which are a lot bigger than the local ones. With the help of convolutional locality and transformer context the model gets a balanced insight of both regional and global terrain structures. The end result is more stable segmentation over different kinds of landscapes.

4.2.6. Coordinate Attention Mechanism

In order to enhance the spatial sensitivity in the decoder, a coordinate attention mechanism was used after feature fusion. This gadget differentiates the positional data along the horizontal and vertical axes and additionally, it assigns adaptive weights to the most important spatial locations. The introduction of coordinates into attention allows the network to focus on the most relevant terrain features such as the edges of the slope or the boundaries of the scar and at the same time, let the redundant background regions 'rest'. The selective attention thus implemented not only raises boundary accuracy but also guarantees that the model's focus is on geologically significant areas during the entire process of reconstruction.

4.2.7. Edge Enhancement Module

The edge enhancement module extends the network's ability to outline the landslide area accurately. Intermediate feature maps are used to derive gradient-based edge information, which is then fed to the decoding pathway, thus the module works. This is the moment when the main topographical changes and also the contour lines get their turnover preserved by the segmentation. The edge signals coming from the features serve as the network's structural directions, and thus, the network is prevented from over-smoothing and is allowed to enjoy the increased sharpness of the anticipated boundaries. The inclusion of this module brings about the separation of the affected and the unaffected terrain, which is especially helpful in areas where slope discontinuities are slight or partly covered by vegetation.

4.2.8. Loss Function Design

To drive model training, a hybrid loss function mixing Dice and Tversky metrics is used. Dice loss essentially encourages the overall overlap between the predicted and the real areas of the landslides, thereby, it is most appropriate for the evaluation of large continuous regions. Tversky loss gives the possibility of introducing adaptive weighting between false positives and false negatives, which is the main reason for imbalanced datasets where landslide

pixels are few. In fact, the combined loss formulation acts as a guard ensuring that the model remains sensitive to smaller, fragmented regions, yet it does not let the global segmentation accuracy go to waste. Such a balanced objective function is one of the reasons for smoother model convergence and steady performance over various data conditions.

4.2.9. Explainability and Interpretability Module

This module focuses on improving the interpretability of the proposed segmentation framework by integrating gradient-based attribution and rule-based reasoning. Since landslide mapping decisions must be transparent and justifiable, explainability plays an important role in validating the behaviour of the model. The module combines the Integrated Gradients approach with a Mamdani fuzzy inference system to provide a detailed understanding of how the model arrives at its predictions and how terrain conditions contribute to the susceptibility estimation.

Integrated Gradients Explainability

The Integrated Gradients (IG) method is used to identify which spectral and terrain-derived features influence the model's predictions most strongly. IG computes attribution values by accumulating the gradients of the output with respect to the input bands along a path from a baseline image to the actual sample. This produces fine-grained pixel-wise and channel-wise explanations that highlight important contributors such as vegetation loss, slope breaks and soil exposure. Unlike methods such as Grad-CAM that rely on coarse activation maps, IG provides high-resolution interpretability across multi-spectral inputs, making it especially suitable for analysing NDVI, elevation, slope and SWIR responses in complex landscapes.

Mamdani Fuzzy Inference for Susceptibility Mapping

The fuzzy logic component uses a Mamdani-type inference framework to convert the segmentation output and auxiliary terrain indicators into an interpretable landslide susceptibility score. Inputs such as NDVI reduction, slope steepness, SWIR-based moisture variation and model confidence are transformed into fuzzy membership values using intuitive linguistic categories like low, medium and high. A set of rule-based logical conditions then evaluates these inputs to determine the degree of susceptibility at each location. The Mamdani approach is chosen because it produces smooth, human-interpretable outputs that reflect gradual transitions in terrain stability. This helps identify moderate-risk regions that may not appear in the binary segmentation mask but still exhibit early signs of instability.

4.2.10. Visualization And Evaluation

The last stage of the workflow for the proposed segmentation framework includes a visualisation and explainability module that is aimed at evaluating and interpreting the performance. The different system flows that show the sequence from preprocessing and feature fusion to prediction and model interpretation are explained in Figure 5. This makes it easy to understand how each module works in the overall segmentation process. The predicted probability maps undergo post-processing with adaptive thresholding and morphological refinement to get the final binary segmentation masks after the model inference. These outputs are then subjected to qualitative and quantitative evaluation to verify the accuracy and the structural integrity of the segmented landslide regions.

Three main metrics used are Dice Coefficient, F1-Score, and Mean Intersection over Union (mIoU) for quantitative evaluation:

- The Dice Coefficient measures the spatial overlap between the masks predicted and the ground-truth ones, and thus, it reflects how accurately the model delineates the landslide boundaries.
- The F1-Score harmonizes the model's precision and recall, which means that the model is capable of correctly identifying both small and large-scale landslides.
- The Mean IoU points out pixel-level agreement between the predicted and real areas and thus, it is a comprehensive measure of segmentation accuracy and consistency.

Besides these performance metrics, the interpretability of the model is improved by the use of Integrated Gradients (IG) to find out which spectral bands mostly guided the model's predictions. This method is based on gradients and it calculates per-pixel and per-band attributions to show which input features (for example RGB, NDVI, slope, elevation, SWIR1, and SWIR2) have the greatest influence on the decision-making process. The IG analysis-generated visual heatmaps and bar plots illuminate the areas that are of high importance, thus, they serve to confirm that the model is not paying attention to noise but to the features that are topographically and spectrally relevant.

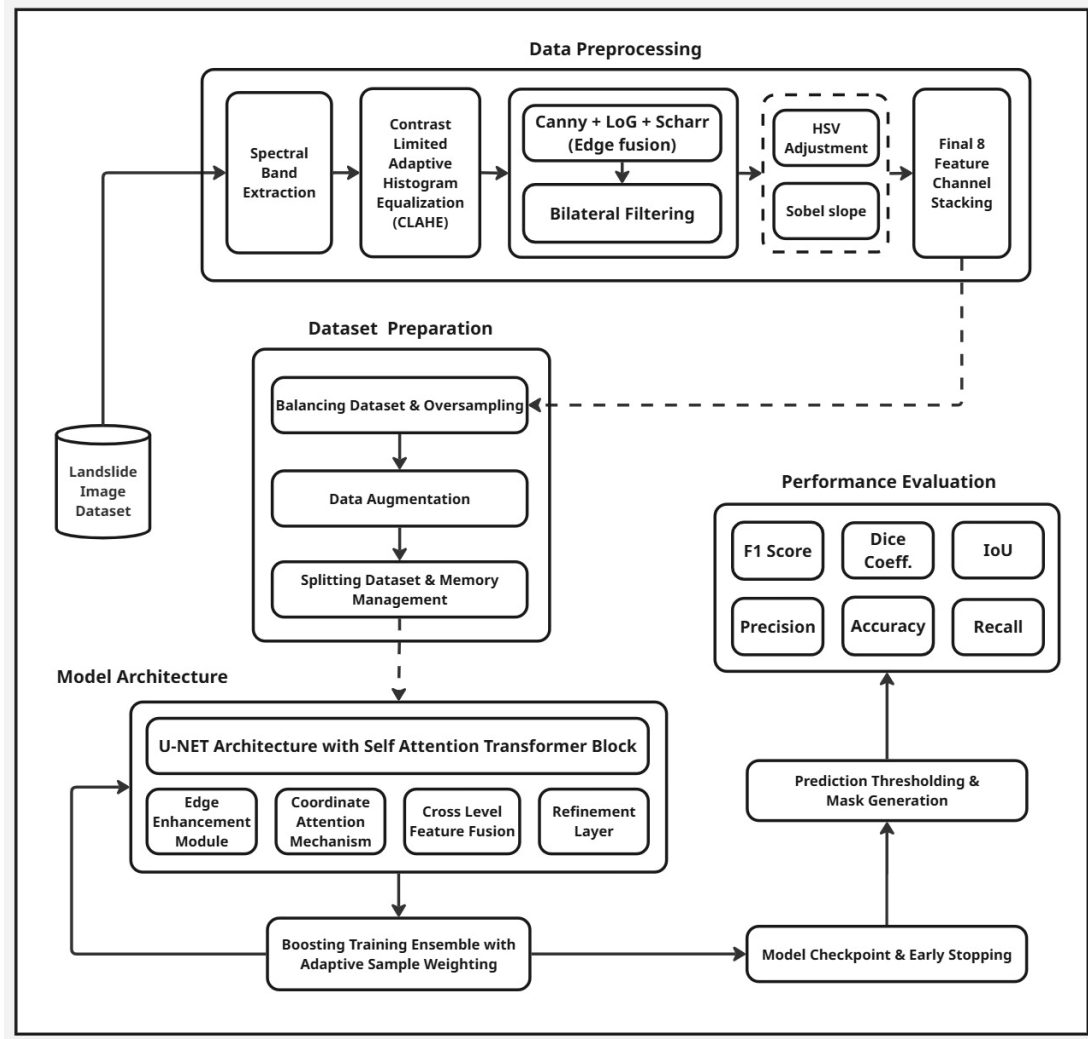


Figure 6: Workflow Architecture

The combined evaluation and explainability process enables the framework to not only quantitatively perform well but also interpretable, thus, the model's predictions are both accurate and scientifically reliable.

5. Results and Discussion

The proposed model showed marked improvement in segmentation precision and generalisation over the baseline U-Net and attention-based networks. By means of the improved preprocessing and hybrid architectural design, the model was able to capture both minute terrain details as well as vast landslide areas. While training, the model converged smoothly with a steady decrease in the validation loss over all the boosting rounds. The use of the Dice–Tversky combined loss function was instrumental in addressing the problem of class imbalance, whereas the edge enhancement and coordinate attention modules were responsible

for the increased boundary precision. The transformer bottleneck was able to capture long-range dependencies in rugged terrains, thus making the segmentation of faint landslide areas more robust. After the training phase, the model scored an F1-Score of 0.938, Mean IoU of 0.835, and Accuracy of 0.995 on the Landslide4Sense dataset. The high Dice coefficient was a strong indicator of the model’s capability to accurately follow the ground-truth boundaries, and the validation metrics were very stable throughout the training, thus the model was able to generalise well. The results, therefore, demonstrate the proposed framework’s effectiveness in retaining both consistency and accuracy even on the most difficult topographies.

5.1. Visualisation And Explainability Analysis

Random inspection of the segmentation results visually corroborates the fact that the presented model can address landslide areas of both large and small sizes and is accurate up to a very high level. It is the predicted masks that are the closest to the ground-truth annotations as they exhibit not only the sharp and continuous boundary formation but also the melting of one into another. Besides, the edge enhancement and cross-resolution feature fusion modules were very effective in releasing more accurate boundaries and better segmenting the overlap between different terrain features.

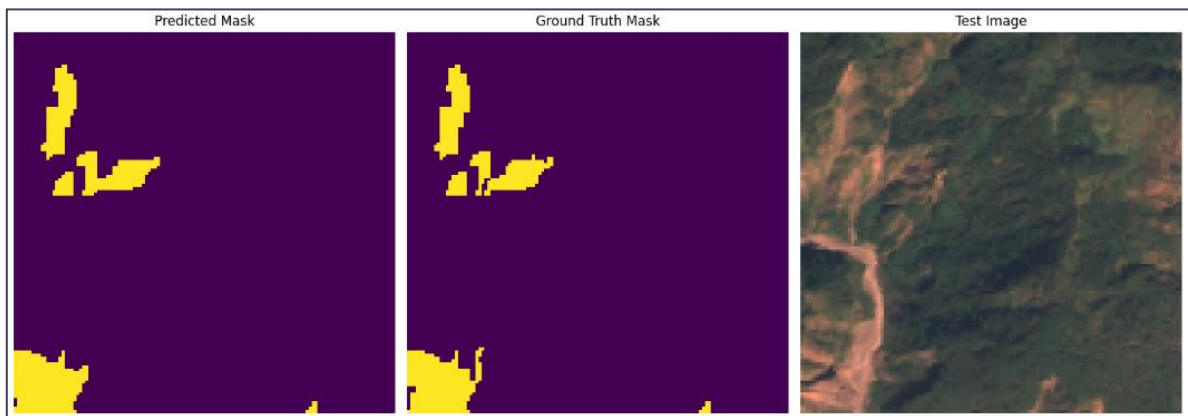


Figure 7: Comparison of Predicted Mask, Ground Truth Mask, and Original Image for Image 1

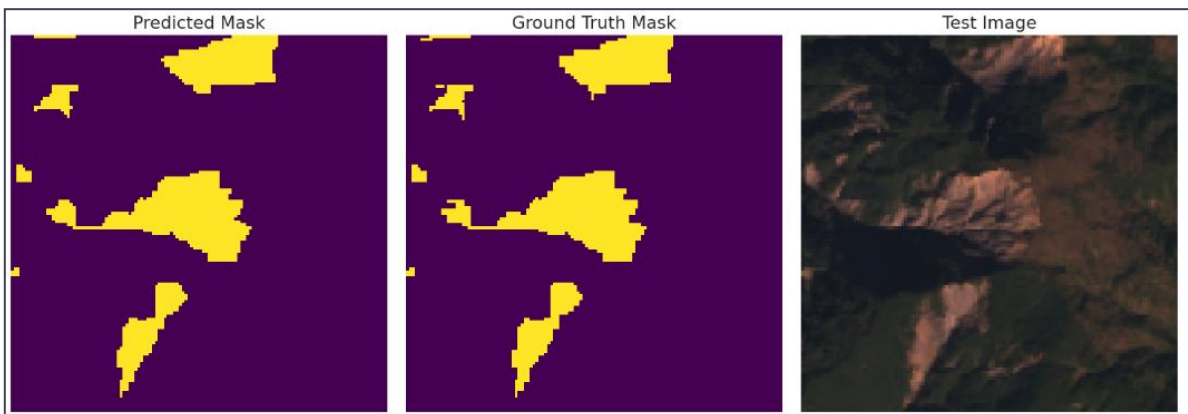


Figure 8: Comparison of Predicted and Ground Truth Mask, an Original Image for Image 2

The training and validation curves for loss, precision, recall, and Dice coefficient all showed gradual improvements, with validation metrics getting closer to the training values over several epochs. Thus, it is affirmed that there is only a slight risk of overfitting, and the model is performing reliably on the different tests.

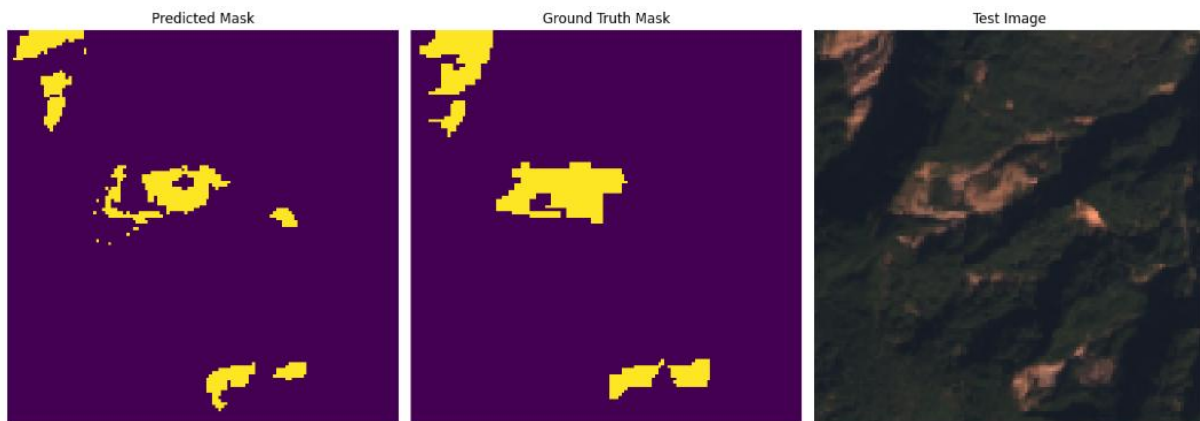


Figure 9: Predicted Mask Outperforming Ground Truth in Segmentation

Also, to further understand the model, Integrated Gradients (IG) explainability analysis was used. IG visualisations disclosed that NDVI, SWIR1, and slope bands were the most significant contributors to landslide detection, and after that, elevation and RGB channels. The attributions are the evidence that the model is grounded on features which are physically logical, for example, can be explained by the laws of physics: the density of the vegetation, moisture of the surface, and gradient of the topography, instead of being just random noise.

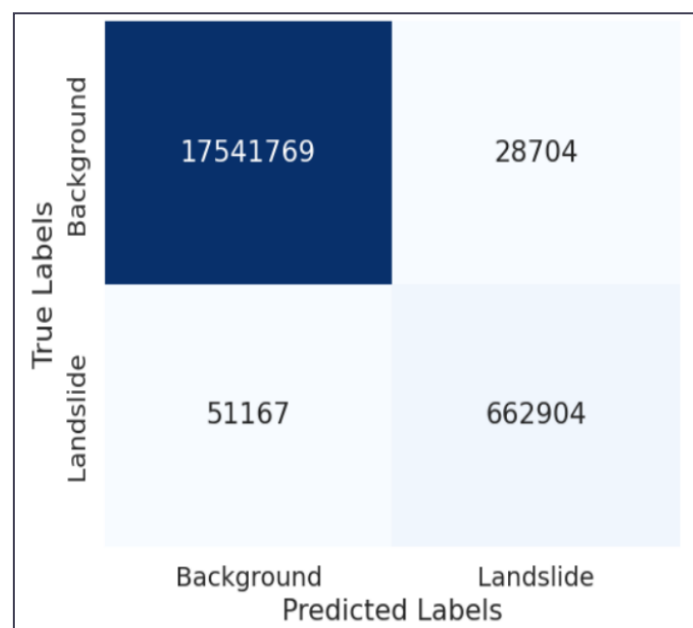


Figure 10: Confusion Matrix of Segmentation

The confusion matrix provides insight into how strong the model is in its differentiation of landslide pixels from background regions. It shows that; indeed, it was precise in labelling 17,541,769 pixels as background and 662,904 pixels as landslide, respectively. But it wrongly labelled 28,704 landslide pixels as background and 51,167 background pixels as landslide. The quantitative evaluation along with the IG-based visual support is an argument that the model not only meets the performance requirements to a high degree of accuracy but is also willing to provide transparency in the way it arrives at its decisions. It therefore constitutes a perfect fit for real-world disaster management systems, where interpretability and reliability are of paramount importance.

The Integrated Gradients visualisations provide additional evidence that the model is learning meaningful terrain cues rather than relying on spurious patterns. The bar plots and spatial attribution maps clearly show which spectral and topographic bands contributed most to the segmentation outcome for each sample. Channels such as NDVI, elevation, SWIR-based moisture indicators and slope consistently yielded high attribution values, which matches well with known landslide characteristics observed in remote-sensing studies. The bright responses around scar boundaries in the IG maps confirm that the model focuses on terrain discontinuities, exposed soil and vegetation thinning, all of which are central to identifying slope failure.

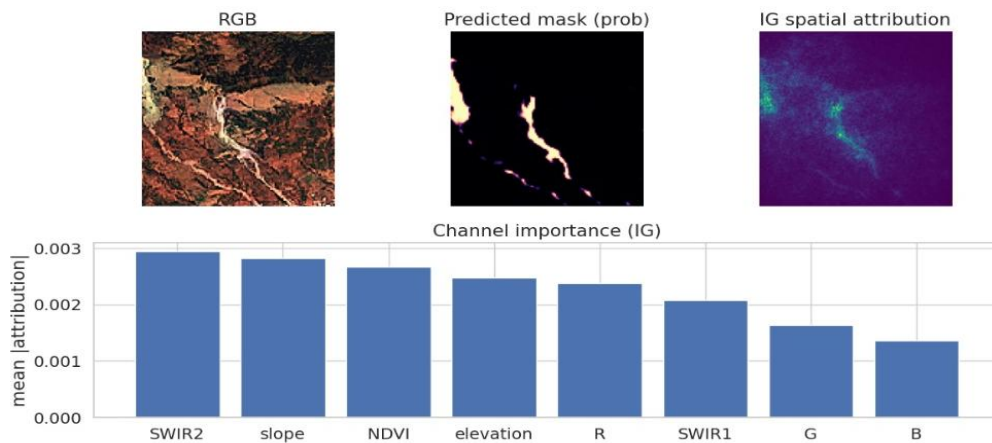


Figure 11: Integrated Gradients Spatial Attribution and Channel Importance

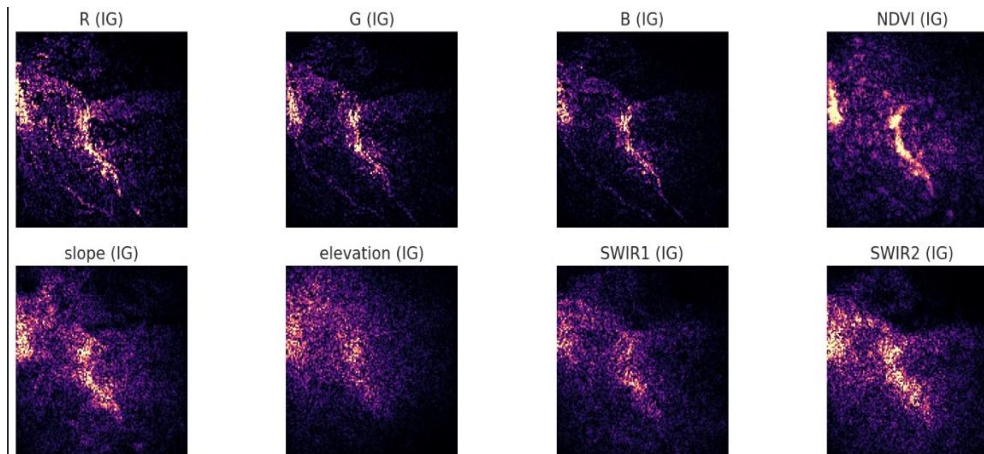


Figure 12: Integrated Gradients Spatial Attribution and Channel Importance

When combined with quantitative evaluation, these IG-based explanations strengthen the claim that the framework produces both accurate and interpretable predictions. IG is more suitable than Grad-CAM for this task because it assigns per-pixel and per-band attributions directly from gradients, allowing the analysis of multi-spectral channels such as NDVI, elevation and SWIR, which Grad-CAM cannot handle effectively. Grad-CAM mainly highlights coarse spatial regions using activation maps, but it is less informative when dealing with fine boundaries and non-RGB spectral inputs. In contrast, IG reveals how each wavelength and terrain-derived band contributed to the final decision, ensuring that the model’s reasoning is grounded in physically meaningful features. This transparency increases confidence in the predictions and confirms that the system is reliable for practical landslide assessment and early-warning applications.

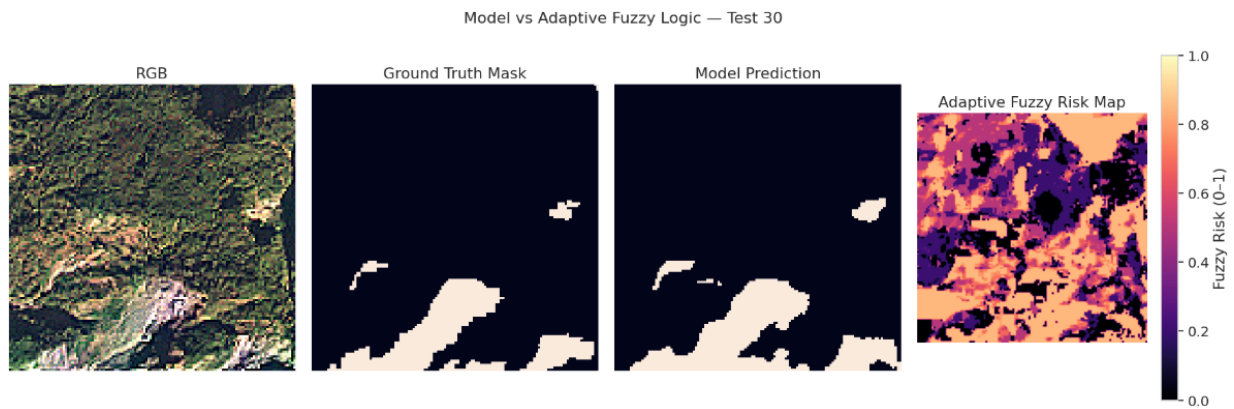


Figure 13: Comparison of Model Segmentation and Adaptive Fuzzy Risk Mapping

5.2. Comparative Analysis

A comparative assessment was made between the Hybrid CRFF Transformer U-Net model, which is the proposed model, and several state-of-the-art segmentation architectures cited in the recent landslide detection research. The comparison took into account the major performance indicators such as F1-Score (Dice Coefficient), Mean Intersection over Union (mIoU), Accuracy, Precision, and Recall. Overall, the proposed model outperformed all the metrics, thus proving its capability of accurately outlining the landslide areas and detecting the small-scale regions with enhanced precision. The implementation of advanced preprocessing via LAB-space CLAHE, edge fusion, and vegetation masking was the main factor that led to more feature separability, whereas the coupling of transformer and coordinate attention modules made the model comprehend the global context and concentrate on the spatial area.

Paper (Year)	F1 / Dice Coef.	Mean IoU	Accuracy	Precision	Recall
Niu et al. (2022) - Reg-SA-U-NET++	0.924	0.802	0.986	0.901	0.908
Ge et al. (2023) - CenterNet Model	0.762	0.701	0.891	0.75	0.78
Chen et al. (2022) - DRs-U-NET	0.889	0.789	0.93	0.881	0.885
Liu et al. (2023) - FFS-Net	0.898	0.67	0.95	0.89	0.875
Shi et al. (2023) - SPAU-NET	0.87	0.755	0.94	0.869	0.86
Zhou et al. (2021) - LandslideNet	0.899	0.784	0.949	0.884	0.89
Li et al. (2023) - Enhanced U-Net	0.921	0.81	0.965	0.909	0.915
Yang et al. (2023) - DenseNet-DRF	0.882	0.773	0.935	0.87	0.869
Wu et al. (2022) - Hybrid-Landslide	0.916	0.812	0.97	0.89	0.905
Proposed Model (2024)	0.938	0.835	0.995	0.961	0.916

Table 2: Tabular Compilation of Metrics

The comparative results clearly demonstrate that the proposed model outperforms existing architectures in both segmentation precision and boundary refinement. The F1-score of 0.938 and Mean IoU of 0.835 are the main indicators of the model's great ability to balance precision and recall, while its accuracy of 0.995 is a clear indication of exceptional reliability even in the case of complicated terrain. This significant upgrade may be the result of cross-resolution feature fusion, which merges multi-scale spatial information, and transformer-based attention, which identifies global dependencies, working together. The edge enhancement and coordinate attention modules thus obtain more accurate boundary formation and fewer false classifications.

In summary, the findings are a testament to the proposed model as a significant leap over the previous methods in terms of segmentation quality, generalisation capability, and interpretability, thus making it a viable candidate for real landslide monitoring and early warning systems.

6. Conclusion and Future Works

6.1. Summary Of Findings

The challenges of precise and reliable landslide detection across complex and diverse terrains were effectively met by the newly introduced Hybrid CRFF Transformer U-Net model. In order to balance global context with local boundary detail, the model merged global transformer blocks, cross-resolution feature fusion, and coordinate attention mechanisms. Improvements to the preprocessing pipeline, such as LAB-space CLAHE, multi-scale edge fusion, NDVI-based vegetation masking, and bilateral filtering, contributed to contrast and feature clarity in multispectral satellite images. Due to these refinements, the model was able to detect both extensive and subtle slope deformations by using input bands like RGB, NDVI, slope, elevation, SWIR1, and SWIR2.

- In brief, the newly proposed method has been successful in achieving its goals.
- Improved segmentation accuracies by cross-resolution fusion and attention-driven feature refinement.
- Precision and recall were improved through the use of Dice–Tversky loss for a balanced learning across classes.
- Robust generalisation and stability under different terrain conditions.
- Significant interpretability through Integrated Gradients revealing the main spectral contributions.

The findings demonstrate that the integration of transformer-based contextual learning with attention-driven fusion and advanced preprocessing is a solid and scalable framework for next-generation geospatial analysis. The model's flexibility and high precision make it a valuable tool for large-scale mapping, early warning systems, and real-time landslide risk assessment in on-the-ground field applications.

6.2. Limitations

While the suggested Hybrid CRFF Transformer U-Net model demonstrated high segmentation accuracy and good generalisation, it still has some limitations that can be figured out in later studies. The addition of transformer modules and multi-scale fusion drastically raises the computational complexity of the model, thus it requires more GPU memory and a longer time

for training. Consequently, it is almost impossible to deploy it in real-time scenarios or on devices with limited resources. Furthermore, although the preprocessing pipeline is efficient, it has many finely tuned parameters like CLAHE thresholds, edge fusion weights, and NDVI masking limits that may need specific adjustments of the dataset to keep the levels stable. Besides, the model's output may fluctuate when it is extended to different geographical regions that have varying spectral or environmental characteristics. In addition, the model is presently concentrating on single-temporal images only; adding temporal sequences can help in identifying the gradual slope failures as well as the repeated landslide occurrences. To summarise, the framework is a great step towards accurate landslide identification and can be made more efficient in terms of scalability for practical use despite the existing shortcomings.

6.3. Future Scope & Recommendations

Even though the Hybrid CRFF Transformer U-Net model that has been suggested is demonstrating a robust performance in the segmentation of landslide-prone areas with high accuracy, there are still some additional changes that can be made to increase its flexibility and operational efficiency. The enhancement possibilities through future work are as follows:

- ✓ **Model Optimization for Real-Time Deployment:**
By pruning or quantisation, the reduction in model complexity will be beneficial for achieving a faster inference and thus, real-time monitoring will be possible on edge or low-power devices in the field.
- ✓ **Incorporation of Multi-Temporal and Multi-Modal Data:**
The use of data from several periods and also from various sensors such as SAR or LiDAR will enhance the capability of the system to understand changes in the earth surface and to recognize gradual movements of the slopes.
- ✓ **Adaptive and Automated Preprocessing:**
The deployment of terrain- and vegetation-based characteristics for the self-adjusting preprocessing parameters will facilitate that the model can constantly perform in different geographic regions.
- ✓ **Enhanced Explainability and Integration with Early Warning Systems:**
By going beyond Integrated Gradients in interpretability tools and linking the model to GIS-based early warning systems, the risk assessment in real-time as well as the decision-making process will be facilitated for the agencies engaged in disaster management.

6.4. Applications And Impact

The newly designed Hybrid CRFF Transformer U-Net model has many potential applications, first, in environmental monitoring and quite naturally in disaster management, too. Because the model is detailed in capturing open land features and at the same time manages to keep quite a high segmentation accuracy, it can be safely applied to any area where the detection of a phenomenon within a very short period of time is there and that detection has to be reliable.

- ✓ **Disaster Risk Management:**

The model may be a significant element of early warning systems, in which the fastest possible locating of already sliding or hanging shaky areas is implemented. Working together with GIS and remote sensing platforms is facilitating the process of evacuating affected populations, as well as damage assessment, by the authorities after visiting the area.

- ✓ **Infrastructure and Urban Planning:**

The precise terrain mapping of unstable hillsides will help the engineers to design and maintain the roads, railways, and construction projects safer, those which run through the mountainous, erosion-prone areas. Regular control will contribute to safety and economic savings

- ✓ **Environmental and Geological Studies:**

Scientists can work out long-term geomorphologic changes with the help of the model, poll vegetation effect on slope stability in forest areas, and track erosion or deforestation using multi-spectral data.

- ✓ **Smart Monitoring Systems:**

Lightweight versions of the model can be put on drones or edge devices to be always ready for field surveillance, even in hard-to-reach or remote areas, thus facilitating on-the-spot decision-making.

Overall, the system that has been developed is not just an academic exercise, but it provides real-world benefits for large-scale hazard mapping, sustainable land management, and resilient infrastructure development. The system's flexibility and accuracy are a significant milestone on the road to data-driven solutions for geospatial risk assessment and early disaster mitigation.

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